

BMI CALCULATOR

Project Documentation

Project Title	BMI Calculator
Technology	Python
Project Type	Command-Line Application
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Education	MCA – GITA Autonomous College (2023–2025)

A beginner-friendly Python command-line application that calculates Body Mass Index (BMI) from user-provided weight and height and classifies the result into standard adult BMI categories.

1. Introduction

The BMI Calculator is a simple command-line application developed using Python. It accepts a user's weight in kilograms and height in meters, calculates BMI using the standard formula, rounds the result to two decimal places, and displays a category.

The project demonstrates core Python programming concepts including user input, data type conversion, arithmetic operations, conditional statements, input validation, exception handling, and formatted output.

2. Project Objective

- Calculate BMI using weight and height supplied by the user.
- Classify the calculated BMI into Underweight, Normal, Overweight, or Obese.
- Handle invalid and non-numeric input safely.
- Round the BMI result to two decimal places.
- Practice fundamental Python programming concepts through a practical project.

3. Problem Statement

Manual BMI calculations require applying the formula and then checking the result against multiple category ranges. This project provides a small automated solution that performs both steps from a command-line interface.

4. Technology Used

Technology / Concept	Purpose
Python	Programming language used to build the application
input()	Collects weight and height from the user
float()	Converts numeric input into decimal values
Arithmetic operators	Calculates BMI
if / elif / else	Classifies BMI
try / except	Handles non-numeric input
round()	Displays BMI to two decimal places

5. Key Features

- Accepts weight in kilograms and height in meters.
- Calculates BMI automatically.
- Rounds BMI to two decimal places.
- Displays a BMI category.
- Rejects zero or negative weight/height values.
- Handles non-numeric input with a clear error message.
- Runs directly in a terminal or command prompt.

6. BMI Formula

BMI = Weight (kg) / Height² (m)

Example: For 60 kg and 1.65 m, BMI = $60 / (1.65^2) = 22.04$, which falls in the Normal category.

7. BMI Categories

BMI Range	Category
Below 18.5	Underweight
18.5 – 24.9	Normal
25.0 – 29.9	Overweight
30.0 and above	Obese

Note: These are simplified adult BMI categories for educational use and are not a medical diagnosis.

8. System Requirements

- Python 3.x
- VS Code or any Python-compatible IDE
- Windows, Linux, or macOS
- Basic terminal/command prompt access

9. Project Structure

```
BMI – Calculator /
|
|-----BMI_Calculator.py
|-----README.md
|-----BMI_Calculator_Project_Documentation.pdf
|-----Screenshots/
|               |
|               |----- Output.png
```

10. Working Process

Start □ Enter Weight □ Enter Height □ Validate Input □ Calculate BMI □ Round BMI □ Determine Category □ Display Result □ End

11. Program Implementation

```
# BMI Calculator Project
Print("==== BMI Calculator ==== ")
while(True):
    try:
        # users input
        weight=float(input("enter your weight in kg :"))
        height=float(input("enter your height in m :"))
        if (weight<=0 or height<=0):
            print("don't enter negative value for weight snd height and try again.....")
        else:
            # calculate BMI
            BMI=weight/(height**2)
            # classify BMI into catagory
            if BMI<18.5:
                catagory="Underweight"
            elif BMI<25:
                catagory="Normal"
            elif BMI<30:
                catagory="Overweight"
            else :
                catagory="Obese"
            print("-----")
            print("\n BMI :",round(BMI,2))
            print("\ncatagory:",catagory)

    except ValueError:
        print("Don't try to enter alpnums, strs and symbols ...invalid input and try again....")
```

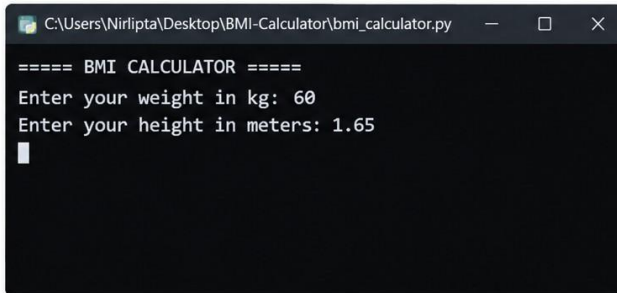
12.Input Validation

The program uses a try-except block to catch ValueError when the user enters non-numeric data. It also checks that weight and height are greater than zero before performing the calculation.

13.Testing

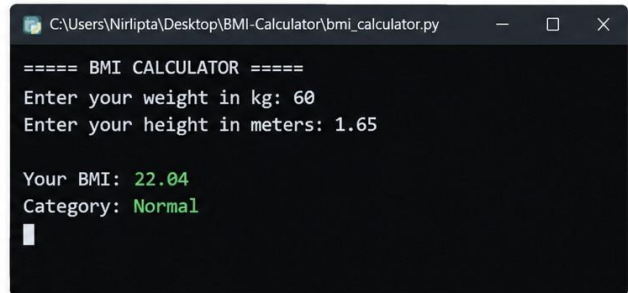
Input	Expected Result
60 kg, 1.65 m	BMI 22.04 – Normal
45 kg, 1.65 m	Underweight
80 kg, 1.65 m	Overweight
100 kg, 1.65 m	Obese
-10 kg, 1.65 m	Invalid positive values message
60 kg, 0 m	Invalid positive values message

12. Project Screenshots



```
C:\Users\Nirlipta\Desktop\BMI-Calculator\bmi_calculator.py
===== BMI CALCULATOR =====
Enter your weight in kg: 60
Enter your height in meters: 1.65
█
```

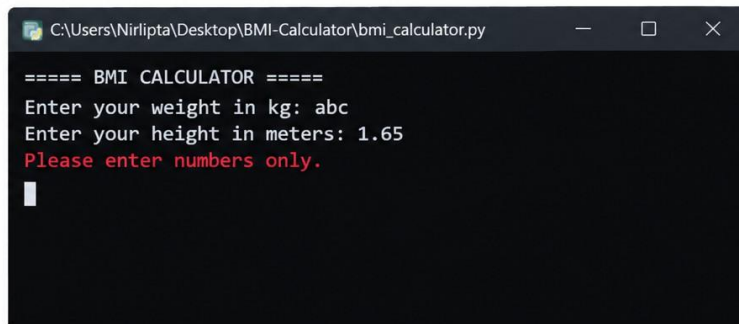
1. BMI Calculator Interface (User Input)



```
C:\Users\Nirlipta\Desktop\BMI-Calculator\bmi_calculator.py
===== BMI CALCULATOR =====
Enter your weight in kg: 60
Enter your height in meters: 1.65

Your BMI: 22.04
Category: Normal
█
```

2. BMI Calculation Result



```
C:\Users\Nirlipta\Desktop\BMI-Calculator\bmi_calculator.py
===== BMI CALCULATOR =====
Enter your weight in kg: abc
Enter your height in meters: 1.65
Please enter numbers only.
█
```

3. Invalid Input Message

The following screenshots demonstrate the calculator interface, a successful BMI calculation, and handling of invalid input.

13. Advantages

- Simple and easy to use.
- Fast BMI calculation.
- Includes basic input validation.
- Demonstrates important Python fundamentals.
- Easy to run without external libraries.

14. Limitations

- Command-line interface only.
- Does not store previous calculations.
- No database or user account system.
- No graphical or web interface.
- Designed for learning and demonstration purposes.

15. Future Enhancements

- Build a graphical user interface.
- Create a web version.
- Store BMI history in a database.
- Add charts and progress tracking.
- Add user profiles and login functionality.

16. Learning Outcomes

This project helped demonstrate practical use of Python input handling, type conversion, arithmetic operations, conditional logic, validation, exception handling, rounding, and command-line application development.

17. Conclusion

The BMI Calculator successfully provides a simple automated way to calculate BMI and classify the result. The project is suitable as a beginner Python project and demonstrates core programming concepts in a practical application.

18. Author

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Full Stack Python Developer

Declaration: This project was developed for educational and learning purposes.